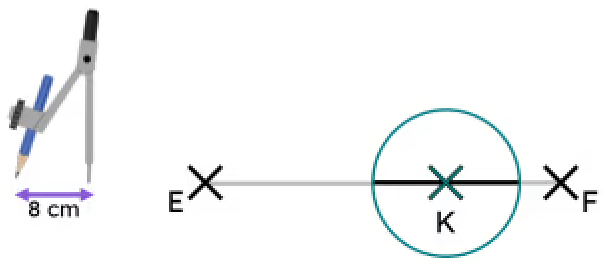
Perpendicular to a given line through a given point

1 The line segment EF has been shortened by constructing a circle of radius 8 cm, with a centre at point K. What is the length (in cm) of the shortened line segment? Fill in the blank



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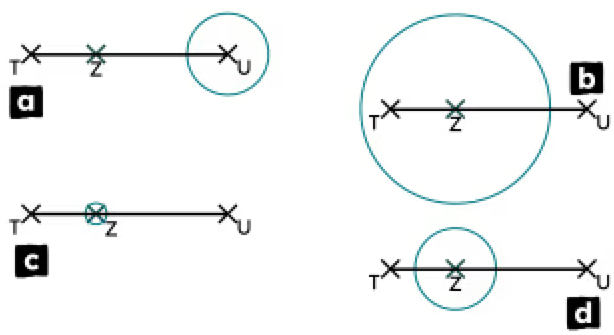
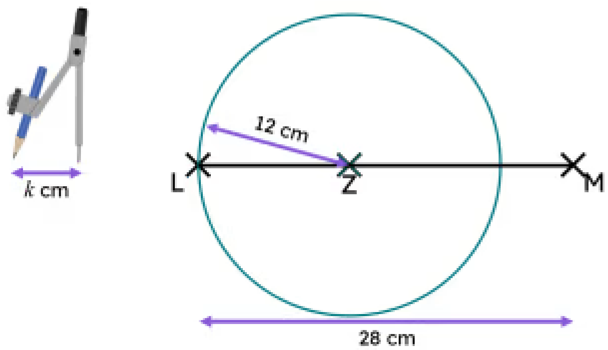
2 Which of these diagrams shows the most appropriate first step when constructing the perpendicular to line segment TU through point Z? Tick **1** correct answer



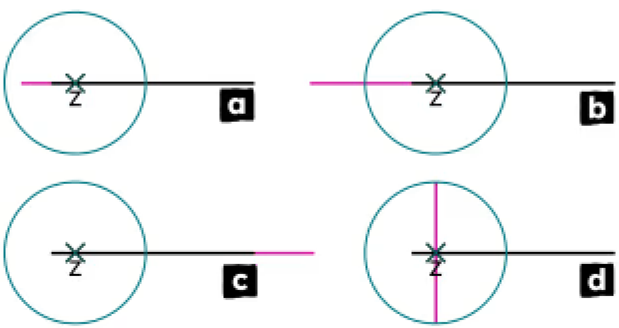
- [ ] a
- [ ] b
- [ ] c
- [x] d

3 Izzy has shortened line segment LM by drawing a circle with centre at point Z. Izzy wants to draw four arcs of radius k cm to construct a perpendicular. What value must k be greater than? Tick **1** correct answer



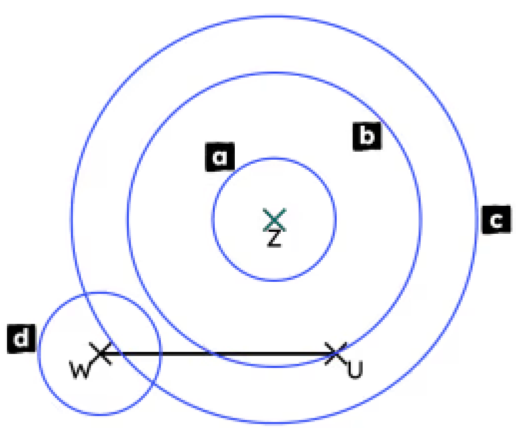
- [ ] 4 cm
- [ ] 6 cm
- [x] 12 cm
- [ ] 14 cm
- [ ] 28 cm

4 A line segment is extended so that a perpendicular can be constructed through point Z. Which of these diagrams shows a suitable extension of the line segment? Tick **1** correct answer



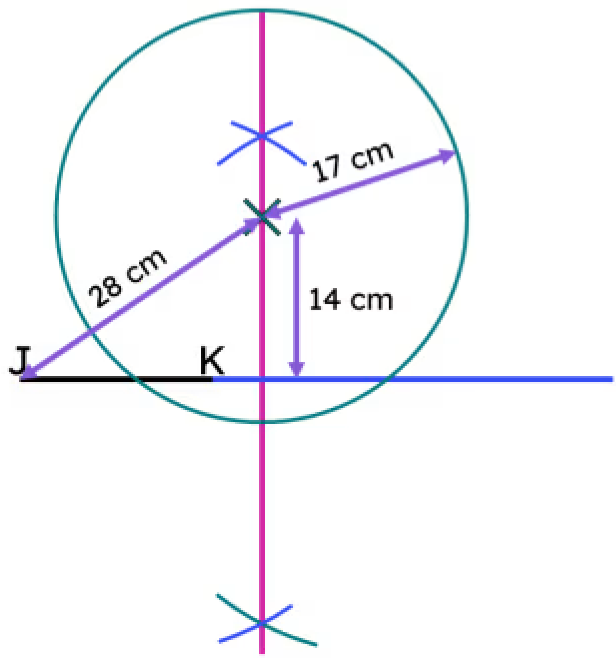
- [ ] a
- [x] b
- [ ] c
- [ ] d

5 Which of these four circles can help in the construction of the perpendicular to line segment WU through point Z? Tick **2** correct answers



- [ ] a
- [x] b
- [x] c
- [ ] d

6 Match the values to their statements. Write the correct letter in each box



a	14 cm
b	15 cm
c	17 cm
d	28 cm

a	the shortest distance from X to an extension of JK
b	the shortest distance from X to line segment JK
c	the radius of circle with centre at point X
d	the shortest distance from X to point J